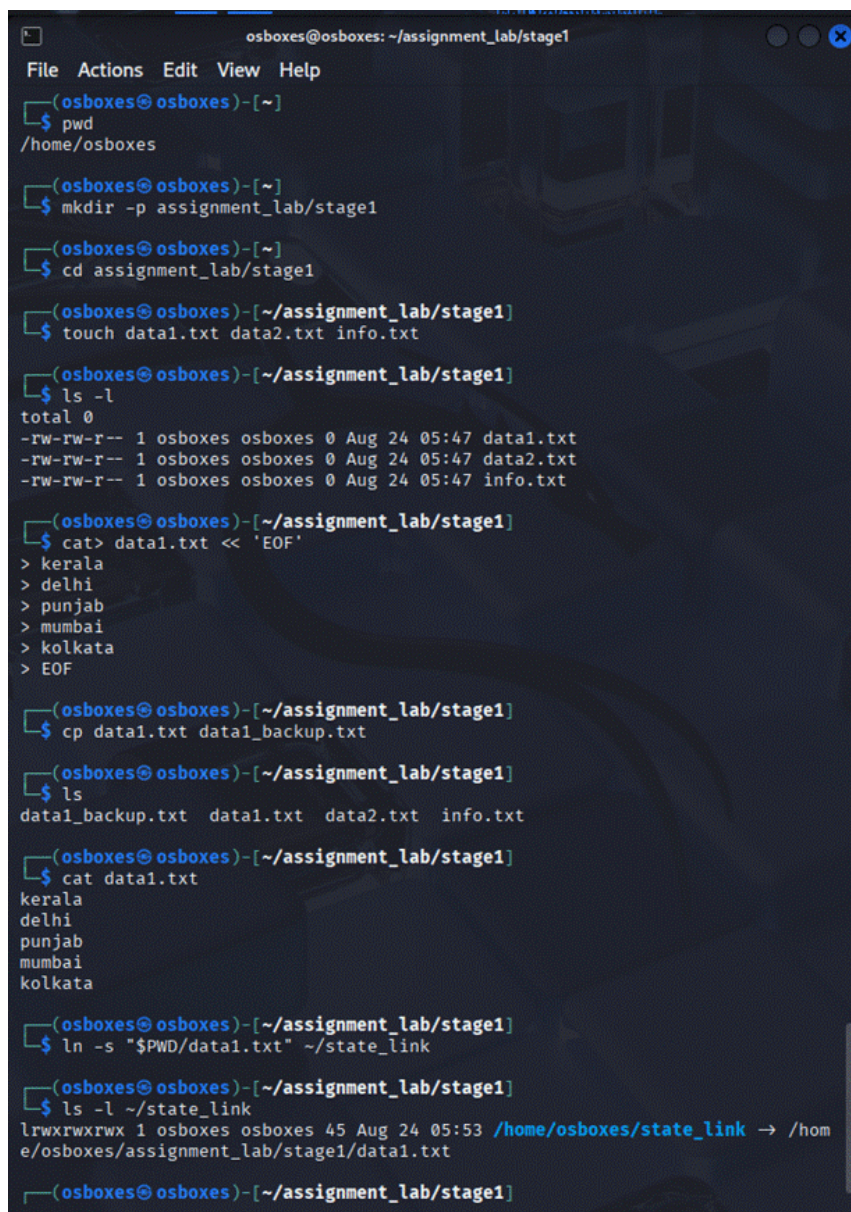


Hands-On Practice Assignment

Practical Lab Tasks

Task 1: Filesystem Structure & File Management

1. Identify your current working directory using `pwd`.
2. Create a new directory structure named `assignment_lab/stage1` using a single command (`mkdir -p`).
3. Navigate into `assignment_lab/stage1` and create three blank text files: `data1.txt`, `data2.txt`, and `info.txt`.
4. Write the names of 5 states into `data1.txt` using `cat > data1.txt` or `echo`.
5. Create a duplicate backup of `data1.txt` named `data1_backup.txt`.
6. Create a soft link on your home directory named `state_link` pointing to `data1.txt`.



```
osboxes@osboxes: ~/assignment_lab/stage1
File Actions Edit View Help
(osboxes@osboxes)-[~]
$ pwd
/home/osboxes
(osboxes@osboxes)-[~]
$ mkdir -p assignment_lab/stage1
(osboxes@osboxes)-[~]
$ cd assignment_lab/stage1
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ touch data1.txt data2.txt info.txt
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ ls -l
total 0
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 data1.txt
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 data2.txt
-rw-rw-r-- 1 osboxes osboxes 0 Aug 24 05:47 info.txt
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ cat > data1.txt << 'EOF'
> kerala
> delhi
> punjab
> mumbai
> kolkata
> EOF
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ cp data1.txt data1_backup.txt
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ ls
data1_backup.txt data1.txt data2.txt info.txt
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ cat data1.txt
kerala
delhi
punjab
mumbai
kolkata
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ ln -s "$PWD/data1.txt" ~/state_link
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ ls -l ~/state_link
lrwxrwxrwx 1 osboxes osboxes 45 Aug 24 05:53 /home/osboxes/state_link -> /home/osboxes/assignment_lab/stage1/data1.txt
(osboxes@osboxes)-[~/assignment_lab/stage1]
```


Task 2: Data Extraction & Sorting

1. Display the contents of data1.txt from top to bottom using cat, and then in reverse order using tac.
2. Sort data1.txt in reverse alphabetical order using sort -r.
3. Search for state names ending with or containing the letter a using grep -i.
4. Count the total words and lines present in data1.txt using wc.

A terminal window titled 'osboxes@osboxes: ~/assignment_lab/stage1' with a menu bar (File, Actions, Edit, View, Help). The terminal shows the following commands and their outputs:

```
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ cat data1.txt
kerala
delhi
punjab
mumbai
kolkata

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ tac data1.txt
kolkata
mumbai
punjab
delhi
kerala

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ sort -r data1.txt
punjab
mumbai
kolkata
kerala
delhi

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ grep i "a" data1.txt
grep: a: No such file or directory
data1.txt:delhi
data1.txt:mumbai

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ grep -i "a" data1.txt
kerala
punjab
mumbai
kolkata

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ grep -i "a$" data1.txt
kerala
kolkata

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ wc data1.txt
  5  5 35 data1.txt

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ wc -l data1.txt
 5 data1.txt

(osboxes@osboxes)-[~/assignment_lab/stage1]
$ wc -w data1.txt
 5 data1.txt
```


Task 3: Archival, Compression & Verification

1. Bundle the entire assignment_lab folder into an archive file named assignment.tar using tar -cvf.
2. Compress assignment.tar using gzip to generate assignment.tar.gz.
3. Use file to verify the compression type of assignment.tar.gz.
4. Preview the decompressed file structure using zcat or tar -tvf.

```
(osboxes@osboxes)-[~/assignment_lab/stage1]
$ cd ~

(osboxes@osboxes)-[~]
$ tar -cvf assignment.tar assignment_lab
assignment_lab/
assignment_lab/stage1/
assignment_lab/stage1/info.txt
assignment_lab/stage1/data2.txt
assignment_lab/stage1/data1.txt
assignment_lab/stage1/data1_backup.txt

(osboxes@osboxes)-[~]
$ ls -lh assignment.tar
-rw-rw-r-- 1 osboxes osboxes 10K Aug 24 06:02 assignment.tar

(osboxes@osboxes)-[~]
$ gzip assignment.tar

(osboxes@osboxes)-[~]
$ ls -lh assignment.tar.gz
-rw-rw-r-- 1 osboxes osboxes 291 Aug 24 06:02 assignment.tar.gz

(osboxes@osboxes)-[~]
$ tar -tvf assignment.tar.gz
drwxrwxr-x osboxes/osboxes 0 2026-08-24 05:46 assignment_lab/
drwxrwxr-x osboxes/osboxes 0 2026-08-24 05:50 assignment_lab/stage1/
-rw-rw-r-- osboxes/osboxes 0 2026-08-24 05:47 assignment_lab/stage1/info.
t
-rw-rw-r-- osboxes/osboxes 0 2026-08-24 05:47 assignment_lab/stage1/data2
xt
-rw-rw-r-- osboxes/osboxes 35 2026-08-24 05:49 assignment_lab/stage1/data1
xt
-rw-rw-r-- osboxes/osboxes 35 2026-08-24 05:50 assignment_lab/stage1/data1
ackup.txt

(osboxes@osboxes)-[~]
$
```

Task 4: System Diagnosis & Remote Monitoring

1. Check your active username with whoami and system uptime with w.
2. Calculate 1542×8 directly from the command line using bc.
3. Verify network reachability to 127.0.0.1 (localhost) sending exactly 3 packets with ping.

```
(osboxes@osboxes)-[~]
$ whoami
osboxes

(osboxes@osboxes)-[~]
$ w
 06:09:48 up  1:07,  1 user,  load average: 0.34, 0.19, 0.11
USER      TTY      FROM          LOGIN@   IDLE   JCPU   PCPU   WHAT
osboxes    -                05:03            0.00s   0.01s lightdm --

(osboxes@osboxes)-[~]
$ echo "1542 * 8" | bc
12336

(osboxes@osboxes)-[~]
$ ping -c 3 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=1.61 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.045 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.100 ms

— 127.0.0.1 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2019ms
rtt min/avg/max/mdev = 0.045/0.584/1.607/0.723 ms

(osboxes@osboxes)-[~]
$ █
```